Amirhosein Mohaddesi

AI/ML Engineer | Robotics & Agentic AI | Neural Networks | PyTorch & ROS2

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Summary

AI/Robotics Engineer with a Ph.D. in Information & Computer Science (Robotics focus) and 5–6 years of experience in ML, multi-agent systems, and simulation. Currently specializing in Agentic AI, integrating Large Language Models (LLMs) with ROS2 to enable autonomous, language-driven decision-making in robotics. Proficient in PyTorch, Hugging Face, and LangChain, with a strong track record in reinforcement learning, transformer models, and emerging agentic AI applications.
• Authorized to work in the US (No sponsorship required)

Research Experience

Graduate Researcher **CA, USA**
CARL Lab, University of California, Irvine *Sep 2021 – Jun 2025*

Platform to train and study navigation agents in ROS2 environment, GitHub

- Built a **multi-robot simulation platform in ROS2** to study human-inspired navigation strategies (*Route, Survey, Mixed*) for team-based exploration and coordination.
- **Scaled the system from Webots to ROS2**, enabling larger swarm sizes, realistic sensor simulation, and integration with SLAM Toolbox, Nav2, and map_merge.
- Conducted **simulation-based experiments** measuring task efficiency, coverage, and cooperation among robot teams with varying navigation strategies.
- Designed a **communication-aware coordination framework** allowing robots to share exploration goals and avoid redundant paths.
- Integrated **C++ and Python nodes** for navigation, mapping, and data logging; visualized multi-robot states in RViz for live analysis.
- Planned extension toward **imitation learning from human trajectory data** and future integration with **LLM-based reasoning** for agent communication and planning.

Benefits of Varying Navigation Strategies in Robot Teams, GitHub

- Investigated the benefits of varying human-inspired navigation strategies (Route, Survey, Mixed) in robot teams through simulation-based experimentation.
- Developed a multi-agent setup in **Webots** using **Clearpath PR2** robots to evaluate task performance and strategy efficiency.
- Implemented obstacle avoidance and conflict resolution algorithms using a **C++** controller.
- Analyzed the impact of navigation strategy on task completion time, environment coverage, and coordination effectiveness.
- Demonstrated that mixed strategies yield a robust balance between exploration and efficiency in team-based navigation tasks.
- Contributed insights applicable to real-world exploration, search, and rescue missions involving autonomous robot teams.

Mohaddesi, S.A.; Hegarty, M.; Chrastil, E.R.; Krichmar, J.L. *"Benefit of Varying Navigation Strategies in Robot Teams."* Proceedings of SAB 2024 (17th International Conference on Simulation of Adaptive Behavior), *Lecture Notes in Computer Science*, Vol. 14993, pp. 63–77, Springer, 2024. DOI: 10.1007/978-3-031-71533-4_5. A related study was also published in IEEE ICDL 2024.

Navigation and Cognitive Load in Telepresence Robots

- Led a study evaluating cognitive load in manual vs. autonomous navigation using telepresence robots in a scavenger-hunt task.
- Developed autonomous navigation features, including real-time SLAM mapping in ROS and a custom GUI using PyQt.
- Designed experimental metrics for cognitive load, spatial awareness, user presence, and task efficiency.
- Conducted user studies to evaluate performance across autonomous and manual navigation modes.
- Concluded that autonomy reduced cognitive burden and improved movement efficiency, memory retention, and usability.

Pan, G.; Weiss, T.; Mohaddesi, S.A.; Szura, J.W.; Krichmar, J.L. *"Navigation and Cognitive Load in Telepresence Robots."* IEEE ICDL 2024, pp. 1–6, DOI: 10.1109/ICDL61372.2024.10644592.

Assistant **CA, USA**
NMI Lab, University of California, Irvine *Jul 2020 – Jul 2021*

8-bit quantization technique for spiking neural networks

- Developed an 8-bit quantized spiking neural network (SNN) using **PyTorch** for power-efficient embedded deployment.
- Introduced a custom quantization technique that reduced energy consumption by **12%–18%**.
- Maintained model accuracy within a **3%–7%** margin, validating efficiency/performance trade-offs.

Selected Projects

AI-Driven Resume Builder (CrewAI Multi-Agent System)

2025

- Built a robust multi-agent automation system with CrewAI, JSON/LaTeX tooling, and custom validators to generate ATS-optimized resumes automatically.
- Engineered a modular pipeline with strict schema checks, LaTeX PDF generation, and 20+ supporting tools; *code available upon request*.

Alter Ego: Personalized Conversational AI, Hugging Face Demo

2025

- Designed and deployed a conversational AI agent that represents my professional background and research expertise.
- Built using **Python, Gradio, and the OpenAI API** with local persona embeddings for adaptive, context-aware conversations.
- Deployed on **Hugging Face Spaces** for public access and website integration.

Multi-Robot Coordination and Distributed Control System, UC Irvine – CARL Lab GitHub

2024

- Built a **ROS2 Humble-based simulation platform** for multi-robot coordination in disaster-response environments.
- Integrated **SLAM Toolbox, Nav2, map_merge, and frontier exploration** for collaborative navigation and mapping.

Benefits of Varying Navigation Strategies in Teams of Robots, UC Irvine – CARL Lab GitHub

2022

- Investigated how **Route, Survey, and Mixed** strategies affect multi-robot team efficiency in **ROS2 and Webots** with Clearpath PR2 robots.

Lunar Lander Trajectory Prediction (LLTP), UC Irvine – PSYCH 239 GitHub

2021

- Built trajectory predictors with **RNNs** and **Convolutional Autoencoders** in **OpenAI Gym**; showed simple RNNs can learn Newtonian motion patterns.

Skills

- **LLMs & Agentic AI**: OpenAI API, LangChain, RAG, Prompt Engineering, LoRA/PEFT
- **Machine Learning**: CNN, RNN, Autoencoders, Transformers, RL (DQN, PPO), VAE, SNNs, Quantization
- **Programming**: Python, C++, Bash **Frameworks**: PyTorch, TensorFlow, OpenCV, HF Transformers, scikit-learn, Keras
- **Data/Viz**: NumPy, Pandas, Matplotlib, Seaborn, SQL, Jupyter **Sim**: ROS2, Gazebo, Webots, OpenAI Gym
- **OS/Tools**: Linux (Ubuntu), Windows, Git, Docker, Conda

Education

Irvine, CA
University of California, Irvine *Ph.D. in Information and Computer Science (ICS)* *Sep 2019 – Jun 2025*

- Cumulative GPA: 3.93
- Relevant Courses: Embedded Ubiquitous Systems, Machine Learning, Neural Networks, Computational Neuroscience, Cognitive Robotics
- Awards: Dean's Award, Donald Bren School of ICS

Tehran, Iran
Sharif University of Technology *B.S. in Computer Engineering* *Sep 2015 – Jun 2019*

- Cumulative GPA: 3.95
- Relevant Courses: Computer Architecture, Embedded Systems, Electrical Circuits, VLSI, Real-Time Processing

Achievements

- **Direct PhD fellowship** – Donald Bren School of Information and Computer Science **UCI, Irvine, CA** *2019*
- **Dean's Award** – Donald Bren School of Information and Computer Science **UCI, Irvine, CA** *2019*
- **Silver Medal** – Iran's National Physics Olympiad, NODET **Tehran, Iran** *2013*